

$$c). V_{Y^2} = E(Y^4) - (E(Y^2))^2 = 0,25 \cdot ((-2)^4 + (-1)^4 + 1^4 + 2^4) - \left(0,25 \cdot ((-2)^2 + (-1)^2 + 1^2 + 2^2) \right)^2 = 8,5 - 6,25 = \underline{\underline{2,25}}$$

Завдання на 14.11.25

2. Approximately 3 percent of all children born in a certain region are from multiple births (that is, twins, triplets, etc.). Of the children born in the region who are from multiple births, 20 percent are left-handed. Of the children born in the region who are from single births, 10 percent are left-handed.

- What is the probability that a randomly selected child born in the region is left-handed?
- What is the probability that a randomly selected child born in the region is a child from a multiple birth, given that the child selected is left-handed?
- A random sample of 20 children born in the region will be selected. What is the probability that the sample will have at least 3 children who are left-handed?
- In the previous question, what is the expectation and variance of the number of left-handed children in the sample?

$$a). \begin{array}{ll} H_1 - \text{multiple births} & P(H_1) = 0,03 \\ H_2 - \text{single births} & P(H_2) = 0,97 \end{array}$$

A. left-handed

$$P(A|H_1) = 0,2$$

$$P(A|H_2) = 0,1$$

$$P(A) = 0,03 \cdot 0,2 + 0,97 \cdot 0,1 = 0,103$$

$$b). P(H_1|A) = \frac{0,03 \cdot 0,2}{0,03 \cdot 0,2 + 0,97 \cdot 0,1} = \frac{0,006}{0,103} = \frac{6}{103}$$

$$c). X - \text{к-во леворуких детей} \quad X \sim \text{Bin}(p = 0,103; n = 20)$$

$$P(X \geq 3) = 1 - (P(X=0) + P(X=1) + P(X=2))$$

$$P(X=0) = 0,103^0 \cdot 0,897^{20} \cdot C_{20}^0$$

$$P(X=1) = 0,103^1 \cdot 0,897^{19} \cdot C_{20}^1$$

$$P(X=2) = 0,103^2 \cdot 0,897^{18} \cdot C_{20}^2$$

d). Ecuem $X \sim \text{Bin}(n; p)$, 10

$$E(X) = n \cdot p$$

$$\text{Var}(X) = n \cdot p \cdot (1-p)$$

$$E(X) = 20 \cdot 0,103 = 2,06$$

$$\text{Var}(X) = 20 \cdot 0,103 \cdot 0,897 = 1,847$$

Imbec: ...

Problems for submission

1. Consider two random variables X, Y with the joint distribution represented by the following table:

X, Y	0	1	2
2	0.3	0.1	0.2
3	0.2	0.1	0.1

$$P(X=2) = 0.3 + 0.1 + 0.2 = 0.6$$

- Find their marginal distributions, $E(X)$, $E(Y)$, $\text{Var}(X)$, $\text{Var}(Y)$.
- Are X and Y independent?
- Find $\text{Cov}(X, Y)$ and $\rho(X, Y)$.
- Find $\text{Cov}(X+2, Y-2X+1)$, $E(X^2 \cdot (Y-1))$.

a)

X	2	3
P	0.6	0.4

Y	0	1	2
P	0.5	0.2	0.3

$\downarrow$ $0.3 + 0.2$ $\downarrow$ $0.1 + 0.1$ $\rightarrow 0.2 + 0.1$

$$E(X) = 2 \cdot 0.6 + 3 \cdot 0.4 = \underline{\underline{2.4}}$$

$$\text{Var}(X) = 2^2 \cdot 0.6 + 3^2 \cdot 0.4 - 2.4^2 = 0.24$$

$$\sigma_X = \sqrt{0.24} = 0.489$$

$$E(Y) = 1 \cdot 0.2 + 2 \cdot 0.3 = \underline{\underline{0.8}}$$

$$\text{Var}(Y) = 1 \cdot 0.2 + 2^2 \cdot 0.3 - 0.8^2 =$$

$$= 1.4 - 0.64 = 0.76$$

$$\sigma_Y = \sqrt{0.76} = 0.87$$

b)

$$P(X = x_i \cap Y = y_j) = P(X = x_i) \cdot P(Y = y_j)$$

$$P(X = 2 \cap Y = 0) = P(X = 2) \cdot P(Y = 0)$$

$$0.3 = 0.6 \cdot 0.5$$



$$P(X = 2 \cap Y = 1) = P(X = 2) \cdot P(Y = 1)$$

$$0.1 \neq 0.6 \cdot 0.2$$

$\Rightarrow X \text{ и } Y \text{ зависимые.}$

c).

Problems for submission

1. Consider two random variables X, Y with the joint distribution represented by the following table:

X, Y	0	1	2
2	0.3	0.1	0.2
3	0.2	0.1	0.1

- Find their marginal distributions, $E(X)$, $E(Y)$, $\text{Var}(X)$, $\text{Var}(Y)$.
- Are X and Y independent?
- Find $\text{Cov}(X, Y)$ and $\rho(X, Y)$.
- Find $\text{Cov}(X + 2, Y - 2X + 1)$, $E(X^2 \cdot (Y - 1))$.

$$\text{Cov}(X, Y) = E(X \cdot Y) - E(X) \cdot E(Y) \quad \Rightarrow$$

$$\begin{aligned} E(X \cdot Y) &= \sum x_i \cdot y_j \cdot P(X = x_i \cap Y = y_j) = 2 \cdot 0 \cdot 0.3 + 2 \cdot 1 \cdot 0.1 + \\ &+ 2 \cdot 2 \cdot 0.2 + 3 \cdot 0 \cdot 0.2 + 3 \cdot 0.1 \cdot 1 + 3 \cdot 0.1 \cdot 2 = \\ &= 0.2 + 0.2 + 0.8 + 0 + 0.3 + 0.6 = 1.9 \end{aligned}$$

$$\Rightarrow 1.9 - 2.4 \cdot 0.8 = -0.02.$$

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \cdot \sigma_Y} = \frac{-0.02}{0.489 \cdot 0.87} = -0.047$$

Problems for submission

1. Consider two random variables X, Y with the joint distribution represented by the following table:

X, Y	0	1	2
2	0.3	0.1	0.2
3	0.2	0.1	0.1

- Find their marginal distributions, $E(X)$, $E(Y)$, $\text{Var}(X)$, $\text{Var}(Y)$.
- Are X and Y independent?
- Find $\text{Cov}(X, Y)$ and $\rho(X, Y)$.
- Find $\text{Cov}(X + 2, Y - 2X + 1)$, $E(X^2 \cdot (Y - 1))$.

Properties of Covariance

1. Covariance is symmetric

$$\text{Cov}(X, Y) = \text{Cov}(Y, X)$$

Proof is obvious

2. Covariance of a variable with itself is simply its variance.

$$\text{Cov}(X, X) = \text{Var}(X)$$

Proof

$$\text{Cov}(X, X) = E[(X - \mu_X) \cdot (X - \mu_X)] = E[(X - \mu_X)^2] = \text{Var}(X)$$

3. Covariance of a variable and a constant is zero.

$$\text{Cov}(X, a) = 0$$

Proof

$\text{Cov}(X, a) = E[(X - E(X))(a - E(a))] = E[(X - E(X))(a - a)] = E(0) = 0$. This property is quite obvious. Might there be a relation between something that changes (X) and something that always takes the same value (a)?

4. Constants can be taken out of the Covariance operator.

$$\text{Cov}(aX, bY) = ab\text{Cov}(X, Y)$$

Proof

$$\text{Cov}(aX, bY) = E[(aX - a\mu_X) \cdot (bY - b\mu_Y)] = E[a(X - \mu_X) \cdot b(Y - \mu_Y)] = abE[(X - \mu_X) \cdot (Y - \mu_Y)] = ab\text{Cov}(X, Y)$$

5. When calculating covariance of sums work with it as if you are multiplying them sums by each other: $(X+Y)(L-W) = XL - XW + YL - YW$.

$$\text{Cov}(X+Y, L-W) = \text{Cov}(X, L) - \text{Cov}(X, W) + \text{Cov}(Y, L) - \text{Cov}(Y, W)$$

Proof

$$\begin{aligned} \text{Cov}(X+Y, L-W) &= E[(X+Y - E(X+Y))(L-W - E(L-W))] = E[(X - EX) + (Y - EY)][(L - EL) - (W - EW)] = \\ &= E[(X - EX)(L - EL)] - E[(X - EX)(W - EW)] + E[(Y - EY)(L - EL)] - E[(Y - EY)(W - EW)] = \\ &= \text{Cov}(X, L) - \text{Cov}(X, W) + \text{Cov}(Y, L) - \text{Cov}(Y, W) \end{aligned}$$

$$(x + y) \cdot (l - w) = x \cdot l - x \cdot w + y \cdot l - y \cdot w$$

(d) Find $\text{Cov}(X + 2, Y - 2X + 1)$, $E(X^2 \cdot (Y - 1))$.

$$\begin{aligned} \bullet \text{Cov}(X + 2, Y - 2X + 1) &= \text{Cov}(X, Y) - \text{Cov}(X, 2X) + \text{Cov}(X, 1) \\ &+ \text{Cov}(2, Y) - \text{Cov}(2, 2X) + \text{Cov}(2, 1) = \\ &= \text{Cov}(X, Y) - 2 \cdot \text{Cov}(X, X) = \text{Cov}(X, Y) - 2 \cdot \text{Var}(X) = \\ &= -0,02 - 2 \cdot 0,24 = -0,5 \end{aligned}$$

$$\bullet E(X^2 \cdot (Y - 1)) = E(X^2 \cdot Y - X^2) = E(X^2 \cdot Y) - E(X^2) =$$

$$= \sum x_i^2 \cdot y_i \cdot P(X = x_i \cap Y = y_i) - E(X^2) \quad \textcircled{=}$$

X	2	3
P	0,6	0,4

X, Y	0	1	2
2	0.3	0.1	0.2
3	0.2	0.1	0.1

$$\begin{aligned} & \textcircled{=} 2^2 \cdot 9 \cdot 0,3 + 2^2 \cdot 1 \cdot 0,1 + 2^2 \cdot 2 \cdot 0,2 + 3^2 \cdot 0 \cdot 0,2 + 3^2 \cdot 1 \cdot 0,1 + \\ & + 3^2 \cdot 2 \cdot 0,1 - \left(4^2 \cdot 0,6 + 3^2 \cdot 0,4 \right) = -8,5 \end{aligned}$$

Quadrat: -8,5

2. Let X, Y, Z be random variables with expectations $\mu_x = \mu_y = 0$, $\mu_z = 1$, and standard deviations $\sigma_x = 2$, $\sigma_y = \sigma_z = 1$. Suppose $\rho(X, Y) = 1/2$, and $X \perp\!\!\!\perp Z$, $Y \perp\!\!\!\perp Z$. Find $\text{Cov}(X - Z, 2X + Y)$.

Независимости

$$\begin{aligned}\text{cov}(x - z, 2x + y) &= \text{cov}(x, 2x) + \text{cov}(x, y) - \text{cov}(z, 2x) - \\ &\quad - \text{cov}(z, y) = 2 \cdot \text{cov}(x, x) + \text{cov}(x, y) - 2 \text{cov}(z, x) - \\ &\quad - \text{cov}(z, y) = 2V_{xx}(x) + \text{cov}(x, y) - 2\underbrace{\text{cov}(z, x)}_{=0} - \underbrace{\text{cov}(z, y)}_{=0} \quad \textcircled{=}\end{aligned}$$

- $\text{Var}(X) = \sigma_x^2 = 2^2 = 4.$

- $\text{cov}(x, y) = \rho \cdot \sigma_x \cdot \sigma_y = \frac{1}{2} \cdot 2 \cdot 1 = 1$.

③ $2.4 \times 1 = 2.4$

Duben: 9.

3. Two dice are rolled. Let X and Y be the number of points on the dice. Define the random variables $U = X + Y$ and $V = X - Y$. Are X and Y uncorrelated? Are they independent?

4			✓
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		u, v										
		-5	-4	-3	-2	-1	0	1	2	3	4	5
$(1,1)$	2	0	0	0	0	0	$1/36$	0	0	0	0	0
	3					$1/36$	0	$1/36$				
	4				$1/36$	0	$1/36$	0	$1/36$			
	5			$1/36$	0	$1/36$	0	$1/36$	0	$1/36$		
	6		$1/36$	0	$1/36$	0	$1/36$	0	$1/36$	0	$1/36$	
	7	$1/36$					0					$1/36$
	8		$1/36$				$1/36$				$1/36$	
	9			$1/36$			0		$1/36$			
	10				$1/36$		$1/36$		$1/36$			
	11					$1/36$	0	$1/36$				
	12						$1/36$					

$$\rho = \frac{\text{cov}(x, y)}{\sigma_x \cdot \sigma_y} = \frac{\text{cov}(u, v)}{\sigma_u \cdot \sigma_v} = \frac{E(u, v) - E(u) \cdot E(v)}{\sigma_u \cdot \sigma_v}$$

$$= \frac{0 - 0 \cdot 7}{\sigma_u \cdot \sigma_v} = 0$$

Answer: ...

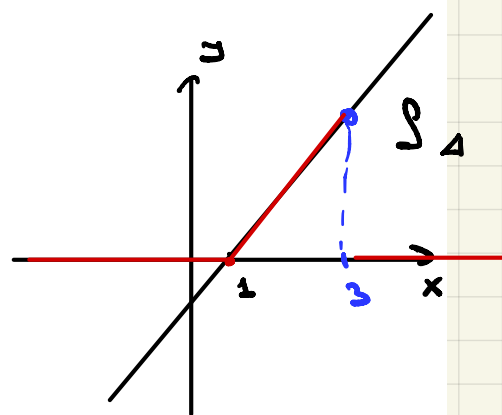
Непрерывная случайная величина.

Задача 2. Случайная величина X задана плотностью вероятности:

$$f(x) = \begin{cases} 0, & x \leq 1; \\ C \cdot (x-1), & 1 < x \leq 3; \\ 0, & x > 3; \end{cases} \quad \alpha=0, \beta=3$$

Требуется:

- а) найти коэффициент C ;
- б) найти функцию распределения $F(x)$;
- в) найти $M(X)$, $D(X)$, $\sigma(X)$
- г) найти вероятность $P(\alpha < X < \beta)$;
- д) построить графики $f(x)$ и $F(x)$.



а). $\int_{-\infty}^{\infty} f(x) dx = 1$

$$S_A = \frac{1}{2} \cdot 2 \cdot h = 1$$

$$\int_1^3 C \cdot (x-1) dx = 1$$

$$\underline{\underline{h = 1}}$$

$$\left. \frac{C \cdot (x-1)^2}{2} \right|_1^3 = 1$$

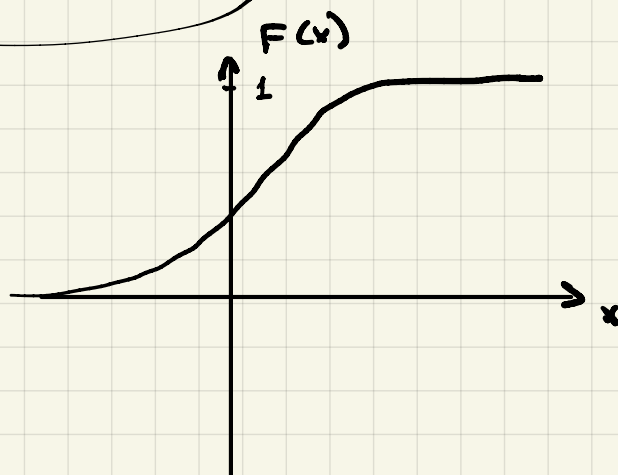
$$\rightarrow C \cdot \left(\frac{x^2}{2} - x \right) \Big|_1^3 =$$

$$\frac{C}{2} \cdot (4 - 0) = 1$$

$$2C = 1$$

$$C = 1/2$$

в). $F(x) = \int_{-\infty}^x f(t) dt$



$$F(t) = \int_1^x \frac{1}{2} \cdot (t-1) dt = \frac{1}{2} \cdot \frac{(t-1)^2}{2} \Big|_1^x = \frac{1}{4} \cdot (t-1)^2 \Big|_1^x$$

$$= \frac{1}{4} \cdot ((x-1)^2 - (1-1)^2) = \frac{(x-1)^2}{4}$$

$$F(x) = \begin{cases} 0, & x \leq 1 \\ \frac{(x-1)^2}{4}, & 1 \leq x < 3 \\ 1, & x \geq 3 \end{cases}$$

$$f(x) = \begin{cases} 0, & x < 1 \\ \frac{1}{2} \cdot (x-1), & 1 \leq x < 3 \\ 0, & x \geq 3 \end{cases}$$

$$3). \quad E(x) = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

$$E(x) = \int_{-\infty}^1 x \cdot 0 dx + \int_1^3 x \cdot \frac{1}{2} (x-1) dx + \int_3^{\infty} x \cdot 0 dx$$

Задача 1.

Дана функция плотности случайной величины $f(x)$:

$$f(x) = \begin{cases} 0, & x \leq 1 \\ c \cdot (x^2 - 1), & 1 < x \leq 4 \end{cases} \quad \text{и } x > 4$$

Найти:

- 1). Константу c
- 2). Математическое ожидание $E(X)$
- 3). Дисперсию $\text{Var}(X)$ и среднее квадратическое отклонение σ
- 4). Определить функцию $F(x)$ и построить график.
- 5). Определить вероятности
 - $P(X = 2)$
 - $P(X > 2)$
 - $P(|X - E(X)| < \sigma)$

1).

$$\int_{-\infty}^{\infty} f(x) dx = 1$$
$$\int_1^4 c \cdot (x^2 - 1) dx = 1$$
$$c \cdot \int_1^4 x^2 - 1 dx = 1$$
$$c \cdot \left(\frac{x^3}{3} \Big|_1^4 - x \Big|_1^4 \right) = 1$$

$$c = \frac{1}{\frac{1}{3} \cdot (4^3 - 1^3) - (4 - 1)} = \frac{1}{21 - 3} = \frac{1}{18}$$

2).

$$\int_1^4 x \cdot f(x) dx = \int_1^4 \frac{1}{18} x \cdot (x^2 - 1) dx = \frac{1}{18} \cdot \int_1^4 (x^3 - x) dx =$$
$$= \frac{1}{18} \cdot \left(\frac{x^4}{4} \Big|_1^4 - \frac{x^2}{2} \Big|_1^4 \right) = \frac{1}{18} \cdot \left(\frac{1}{4} \cdot (4^4 - 1) - \frac{1}{2} \cdot (4^2 - 1) \right) =$$

$4^3 - 0,25 - 15/2$

$$= \frac{1}{18} \cdot \left(\frac{255}{4} - \frac{15}{2} \right) = 3,125$$

$$3). \text{Var}(x) = \int_{-\infty}^{\infty} x^2 f(x) dx - (E(x))^2 =$$

$$= \int_1^4 \frac{1}{18} x^2 \cdot (x^2 - 1) dx - 3,125^2 = \frac{1}{18} \cdot \left(\frac{x^5}{5} \Big|_1^4 - \frac{x^3}{3} \Big|_1^4 \right) - 3,125^2 =$$

$$= \frac{1}{18} \cdot \left(\frac{1}{5} \cdot (1024 - 1) - \frac{1}{3} \cdot (64 - 1) \right) - 3,125^2 = \underline{\underline{0,434}}$$

$$\sigma = \sqrt{\text{Var}(x)} = \sqrt{0,434} = \underline{\underline{0,659}}$$

4). Определить функцию $F(x)$ и построить график.

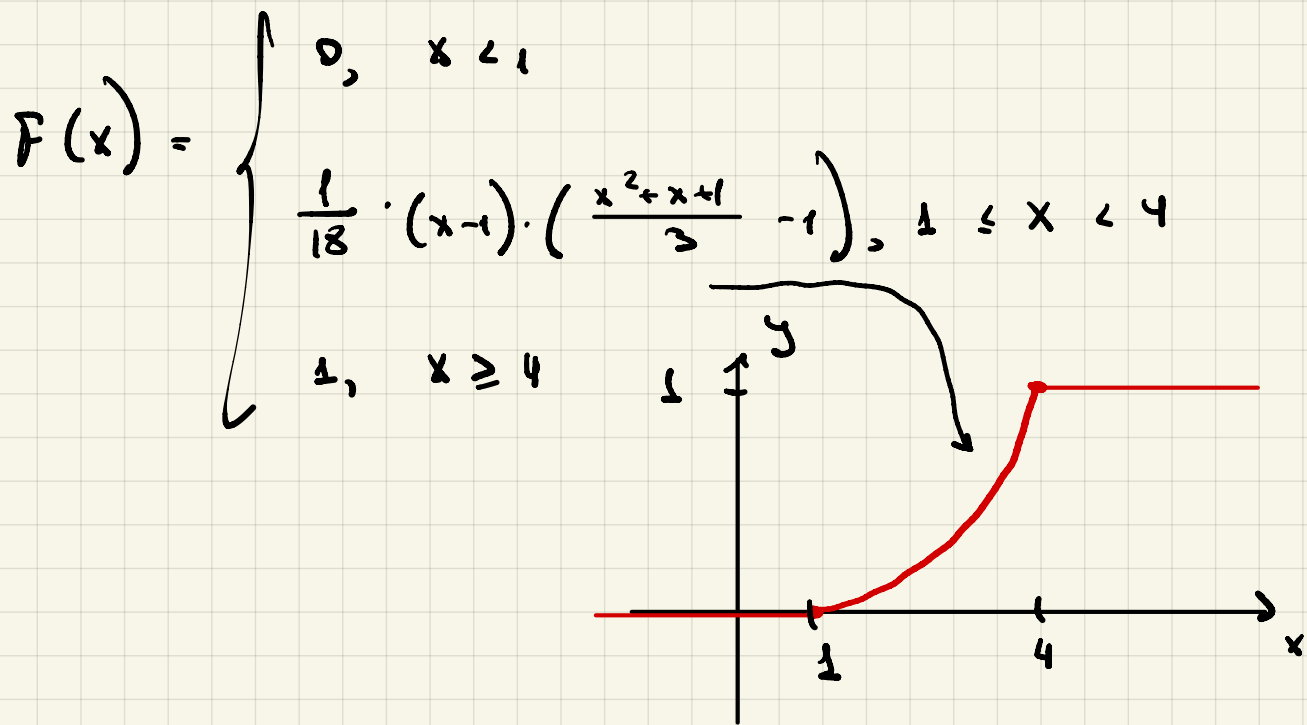
5). Определить вероятности

- $P(X = 2)$
- $P(X > 2)$
- $P(|X - E(X)| < \sigma)$

$$F(x) = \int_{-\infty}^x f(t) dt = \int_1^x \frac{1}{18} \cdot (t^2 - 1) dt = \frac{1}{18} \cdot \left(\frac{t^3}{3} \Big|_1^x - t \Big|_1^x \right) =$$

$$= \frac{1}{18} \cdot \left(\frac{1}{3} \cdot (x^3 - 1) - (x - 1) \right) = \frac{1}{18} \cdot \left(\frac{1}{3} \cdot (x - 1) \cdot (x^2 + x + 1) - (x - 1) \right)$$

$$= \frac{1}{18} \cdot (x - 1) \cdot \left(\frac{1}{3} \cdot (x^2 + x + 1) - 1 \right) = \underline{\underline{\frac{1}{18} \cdot (x - 1) \cdot \left(\frac{x^2 + x + 1}{3} - 1 \right)}}$$



4). Определить функцию $F(x)$ и построить график.

5). Определить вероятности

- $P(X = 2) = 0$
- $P(X > 2)$
- $P(|X - E(X)| < \sigma)$

$$F(x) = P(X \leq x)$$

$$\bullet P(X > 2) = 1 - P(X \leq 2) = 1 - F(2) =$$

$$= 1 - \frac{1}{18} \cdot (2-1) \cdot \left(\frac{2^2+2+1}{3} - 1 \right) = 0,925$$

$$P(X > 5) = P(5 < X < \infty) = F(\infty) - F(5) = 1 - 1 = 0$$

$$\bullet P(|X - E(X)| < 2) = ?$$

$$E(X) = 3,125$$

$$\sigma = 0,659$$



$$-2 < X - E(X) < 2$$

$$E(X) - 2 < X < E(X) + 2$$

$$P(E(x) - 2 \leq x \leq E(x) + 2) = P(3,125 - 0,659 \leq x \leq 3,125 + 0,659) \\ = F(2) - F(0)$$

Problem: ...

Задача 2.

c.d.f

Дана функция распределения случайной величины $F(x)$:

$$F(x) = \begin{cases} 0, & x < 2 \\ \frac{1}{16} \cdot (x-2)^2, & 2 \leq x < 6 \\ 1, & x \geq 6 \end{cases}$$

Найти:

- 1). Математическое ожидание $E(X)$
- 2). Дисперсию $\text{Var}(X)$ и среднее квадратическое отклонение σ
- 3). Определить функцию $f(x)$ и построить график.
- 4). Определить вероятности
 - $P(X = 4)$
 - $P(X > 4)$
 - $P(|X - E(X)| > 1)$
 - $P(|X - E(X)| < \sigma)$

$$1). \quad E(x) = \int_{-\infty}^{\infty} x \cdot f(x) dx$$

$$f(x) = F'(x) = \begin{cases} 0, & x < 2 \\ \left(\frac{1}{16} \cdot (x-2)^2 \right)', & 2 \leq x < 6 \\ 1, & x \geq 6 \end{cases} \quad \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} 0, & x < 2 \\ \frac{x-2}{8}, & 2 \leq x < 6 \\ 0, & x \geq 6 \end{cases} \quad \Leftrightarrow \begin{cases} \frac{x-2}{8}, & 2 \leq x < 6 \\ 0, & \text{otherwise} \end{cases}$$

$$E(x) = \int_2^6 x \cdot \frac{x-2}{8} dx = \frac{1}{8} \int_2^6 (x^2 - 2x) dx =$$

$$= \frac{1}{8} \left(\frac{x^3}{3} \Big|_2^6 - \frac{2x^2}{2} \Big|_2^6 \right) = \frac{1}{8} \cdot \left(\frac{1}{3} \cdot (6^3 - 2^3) - (6^2 - 2^2) \right) = \frac{14}{3}$$

$$2) \quad V_{02}(x) = \underbrace{E(x^2)}_{\int x^2 \cdot f(x)} - (E(x))^2 = \int_2^6 x^2 \cdot \frac{x-2}{8} dx - \left(\frac{14}{3} \right)^2 =$$

$$= \frac{1}{8} \cdot \left(\frac{x^4}{4} \Big|_2^6 - 2 \cdot \frac{x^3}{3} \Big|_2^6 \right) - \left(\frac{14}{3} \right)^2 =$$

$$= \frac{1}{8} \cdot \left(\frac{1}{4} \cdot (36^2 - 2^4) - \frac{2}{3} \cdot (6^3 - 2^3) \right) - \left(\frac{14}{3} \right)^2 =$$

$$= \frac{68}{3} - \left(\frac{14}{3} \right)^2 = \frac{8}{3}$$

+

$$\sigma = \frac{\sqrt{8}}{3} = \frac{2\sqrt{2}}{3}$$

Задача 3.

Дана функция плотности случайной величины $f(x)$:

$$f(x) = \begin{cases} 0, & x < -1 \\ 2a, & -1 \leq x \leq 1 \\ a, & 1 \leq x < 4 \\ 0, & x > 4 \end{cases}$$

Найти значение a и построить график плотности распределения. Найти функцию распределения вероятностей $F(x)$ и построить её график. Вычислить вероятность $P(0 < X < 2)$.

$$1). \int_{-\infty}^{\infty} f(x) dx = \int_{-1}^1 2a dx + \int_1^4 a dx = 2a x \Big|_{-1}^1 + a \cdot x \Big|_1^4 =$$

$$= 2a(1 + 1) + a \cdot (4 - 1) = 4a + 3a = 7a = 1$$

$$a = 1/7$$

$$2). F(x) = P(X \leq x) = \int_{-\infty}^x f(t) dt$$

1. Для $x < -1$:

$$F(x) = 0$$

2. Для $-1 \leq x \leq 1$:

$$F(x) = \int_{-1}^x 2 \cdot \frac{1}{7} dt = \frac{2}{7} t \Big|_{-1}^x = \frac{2}{7} \cdot (x + 1)$$

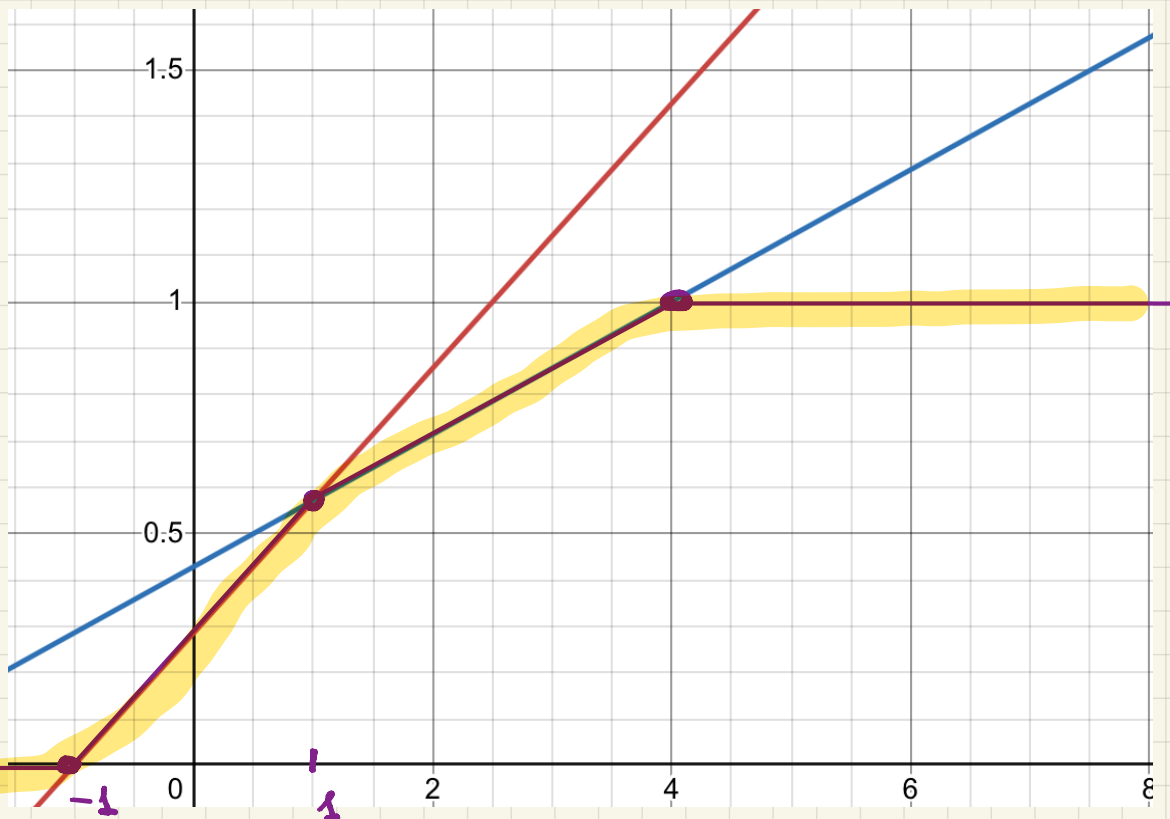
3. Для $1 \leq x \leq 4$:

$$f(x) = \begin{cases} 0, & x < -1 \\ 2a, & -1 \leq x \leq 1 \\ a, & 1 \leq x < 4 \\ 0, & x > 4 \end{cases}$$

$$F(x) = \int_{-1}^1 2 \cdot \frac{1}{7} dt + \int_1^x \frac{1}{7} dt = \frac{2}{7} \cdot (1+1) + \frac{1}{7} \cdot (x-1) =$$

$$= \frac{4}{7} + \frac{1}{7} \cdot x - \frac{1}{7} = \frac{1}{7} x + \frac{3}{7}$$

$$F(x) = \begin{cases} 0, & x < -1 \\ \frac{2}{7} \cdot (x+1), & -1 \leq x \leq 1 \\ \frac{x+3}{7}, & 1 \leq x < 4 \\ 1, & x \geq 4 \end{cases}$$

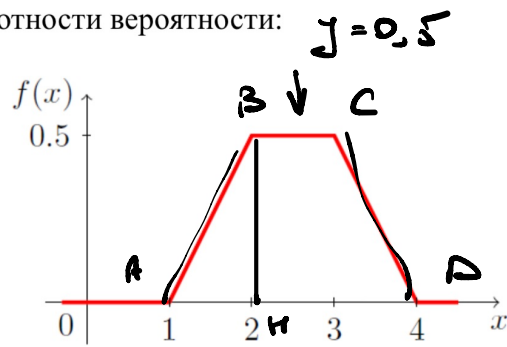


Bei ungen 3 yk-w:

$$F(x) = \int_a^b f(x) dx + \int_b^a f(x) dx + \int_a^x f(t) dt$$

Задача 5.

Рассмотрим непрерывную случайную величину X , которая имеет следующую функцию плотности вероятности:



$$\frac{x - x_A}{x_B - x_A} = \frac{y - y_A}{y_B - y_A}$$

а) Покажите, что общая вероятность равна 1.

б) Найдите $P(X \leq 2)$, $P(2 \leq X \leq 3)$.

в) Нарисуйте график кумулятивной функции (функции распределения) распределения X .

а). $\int_{-\infty}^{\infty} f(x) dx = S$

$$\int_{ABCD} = \frac{1}{2} \cdot (BC + AD) \cdot BH =$$
$$= \frac{1}{2} \cdot (1 + 3) \cdot 0.5 = 1.$$

б). $P(X \leq 2) = S_{\Delta} = \frac{1}{2} \cdot 1 \cdot 0.5 = \frac{1}{4}$